ISRO 2025 Practice Questions - Based on 2023 Pattern

Instructions: These questions follow the exact pattern and difficulty level of ISRO 2023 paper. Each question has 4 options (a,b,c,d). Focus on understanding the approach rather than just memorizing answers.

GRAPH ALGORITHMS & DATA STRUCTURES (Questions 1-8)

1. Find the minimum spanning tree weight for the following graph:
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Graph with vertices A,B,C,D,E and edges:
A-B: 4, A-C: 2, B-C: 1, B-D: 5, C-D: 8, C-E: 10, D-E: 2, B-E: 3
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(a) 8, 4 edges
(b) 10, 4 edges
(c) 12, 4 edges
(d) 15, 5 edges

2. Match the following algorithms with their applications:
(A) Prim's Algorithm (i) Single source shortest path with negative weights
(B) Kruskal's Algorithm (ii) All pairs shortest path
(C) Bellman-Ford (iii) Minimum spanning tree (edge-based)
(D) Warshall Algorithm (iv) Minimum spanning tree (vertex-based)

(a) (A)-(iv), (B)-(iii), (C)-(i), (D)-(ii)
(b) (A)-(iii), (B)-(iv), (C)-(ii), (D)-(i)
(c) (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)
(d) (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)

3. Time complexity to find the height of a binary tree with n nodes is:
(a) O(n^2)
(b) O(n log n)
(c) O(n)
(d) O(log n)

4. The preorder traversal of a binary search tree is (20, 10, 5, 15, 30, 25, 35). The postorder traversal is:
(a) (5, 15, 10, 25, 35, 30, 20)
(b) (35, 30, 25, 20, 15, 10, 5)
(c) (5, 10, 15, 20, 25, 30, 35)
(d) (5, 15, 10, 35, 25, 30, 20)

5. The minimum number of arithmetic operations required to evaluate $f(x,y,z) = x^2 + 2xy + y^2 + z$:
(a) 4
(b) 5
(c) 6
(d) 3

6. Maximum number of simple graphs possible with n vertices:
(a) 2^(n(n-1)/2)
(b) 2^(n-1)/2
(c) 2^(n(n+1)/2)
(d) 2^n(n+1)

7. What is the inorder predecessor of 25 in the BST with preorder: (30, 20, 10, 25, 40, 35, 50):
(a) 20
(b) 10
(c) 30
(d) 35

8. In a priority queue, insertion and deletion can be done at:
(a) Front only
(b) Rear only
(c) Middle only
(d) Any position

THEORY OF COMPUTATION (Questions 9-14)

9. Consider the NFA with ε-transitions for lexical token ID. Which regular expression corresponds to token ID that starts with a letter:
(a) letter (letter | digit)*
(b) (letter | digit)* letter
(c) digit (letter | digit)*
(d) letter (letter* | digit*)

10. Consider the context-free grammar: E → E+E | E*E | id. The string "id + id * id" has:
(a) No parse tree according to G
(b) Exactly one parse tree according to G
(c) Exactly two parse trees according to G
(d) More than two parse trees according to G

11. In the context of parsing techniques, LR(1) refers to:

(a) Left to right scan, rightmost derivation, 1 symbol lookahead
(b) Left to right scan, leftmost derivation, 1 symbol lookahead
(c) Right to left scan, leftmost derivation, 1 symbol lookahead
(d) Left to right scan, leftmost reduction, 1 symbol lookahead

12. Which sequence best describes the compiler phases order:
(a) Lexical → Syntax → Semantic → Code optimization → Intermediate code generation
(b) Syntax → Lexical → Semantic → Intermediate code generation → Code optimization
(c) Lexical → Syntax → Semantic → Intermediate code generation → Code optimization
(d) Lexical → Semantic → Syntax → Code optimization → Intermediate code generation

13. Consider the DFA with alphabet {a,b}. States q0(initial), q1, q2(final). Transitions: δ(q0,a)=q1, δ(q0,b)=q0, δ(q1,a)=q2, δ(q1,b)=q0, δ(q2,a)=q2, δ(q2,b)=q2. The language accepted is:
(a) Strings with even number of a's
(b) Strings ending with aa
(c) Strings containing aa
(d) Strings with odd number of a's

14. The context-free grammar S → aSb | T, T → bT | ε generates the language:
(a) $\{a^nb^m : n \geq m \geq 0\}$
(b) $\{a^nb^m : m > n \geq 0\}$
(c) $\{a^nb^m : n < m\}$
(d) $\{a^nb^n : n \geq 0\}$

OPERATING SYSTEMS (Questions 15-22)

15. Which heap memory allocation strategy is likely to exploit spatial locality best:
(a) Best Fit
(b) First Fit
(c) Next Fit
(d) Worst Fit

16. Which multithreading model is followed in Linux OS:
(a) One-to-One (User thread to Kernel thread)
(b) Many-to-One (Many User threads to One Kernel thread)
(c) One-to-Many (One User thread to Many Kernel threads)
(d) Many-to-Many (Many User threads to Many Kernel threads)

17. Purpose of priority inheritance protocol in synchronization mechanisms:
(a) Prevent priority inversion in systems with multiple priorities
(b) Provide mutual exclusion between threads

(c) Prevent priority inversion and ensure fairness in resource allocation
(d) Allow multiple threads simultaneous resource access

18. Which of the following assigns segment numbers for code and data segments:
(a) Compiler
(b) Assembler
(c) Loader
(d) Linker

19. Match the process schedulers with their activities:
(A) Long term scheduler (i) Executes faster to reduce CPU wastage
(B) Medium term scheduler (ii) Controls degree of multiprogramming
(C) Short term scheduler (iii) Associated with swapping

(a) (A)-(ii), (B)-(iii), (C)-(i)
(b) (A)-(i), (B)-(ii), (C)-(iii)
(c) (A)-(iii), (B)-(i), (C)-(ii)
(d) (A)-(ii), (B)-(i), (C)-(iii)

20. In producer-consumer problem, most appropriate synchronization primitive for consumer to wait when buffer is empty:
(a) Spinlock
(b) Mutex lock
(c) Semaphore
(d) Monitor

21. Any attempt by a process to access memory allocated to OS results into:
(a) Trap to OS
(b) Context switching
(c) Page fault
(d) Scheduler dispatch

22. Which statement about interrupts is FALSE:
(a) Can be triggered by hardware or software
(b) Hardware interrupts triggered via system bus
(c) Software interrupts triggered by system calls
(d) Trap is a hardware generated interrupt

COMPUTER NETWORKS (Questions 23-28)

23. A 1 Gbps link with 50ms RTT uses 2000 byte packets. For >95% channel utilization, minimum window size should be:

(a) 1500 packets
(b) 2500 packets
(c) 3000 packets
(d) 4000 packets

24. In CSMA/CD network, minimum frame size is 2000 bytes, cable length 2km, signal speed 200000 km/sec. Data transfer speed will be:
(a) 800 Mbps
(b) 1.6 Gbps
(c) 500 Mbps
(d) 1 Gbps

25. For 5-layer protocol with applications generating 400 byte messages, each layer adds 8-byte header. Bandwidth waste percentage:
(a) 10%
(b) 12%
(c) 15%
(d) 8%

26. Match 802.11 versions with speeds:
(i) 802.11n   (P) 150 Mbps
(ii) 802.11g   (Q) 11 Mbps
(iii) 802.11b (R) 54 Mbps
(iv) 802.11a   (S) 600 Mbps

(a) (i)-(S), (ii)-(R), (iii)-(Q), (iv)-(R)
(b) (i)-(P), (ii)-(R), (iii)-(Q), (iv)-(R)
(c) (i)-(S), (ii)-(R), (iii)-(Q), (iv)-(P)
(d) (i)-(R), (ii)-(S), (iii)-(Q), (iv)-(P)

27. Which is true about error detection/correction:
(a) Both Parity and CRC are error correcting codes
(b) Both Parity and Hamming codes are error correcting
(c) Both Reed-Solomon and LDPC are error correcting codes
(d) Both CRC and LDPC are error correcting codes

28. In Reverse Polish notation, expression P*Q+R*S is written as:
(a) PQ*RS*+
(b) P*QR*S+
(c) PQ*RS+*
(d) P*Q*RS+

COMPUTER ARCHITECTURE (Questions 29-35)

29. In a vectored interrupt:
(a) Branch address assigned to fixed memory location
(b) Interrupting source supplies branch information to processor
(c) Branch address obtained from processor register
(d) None of the above

30. For given finite instructions, which processor architecture provides fastest execution:
(a) VLIW - Very Long Instruction Word
(b) Super-scalar
(c) RISC - Reduced Instruction Set Computer
(d) All provide same performance

31. An instruction pipeline can be implemented by:
(a) LIFO buffer
(b) FIFO buffer
(c) Stack
(d) None of the above

32. Operation executed on data stored in registers is called:
(a) Macro-operation
(b) Micro-operation
(c) Bit-operation
(d) Byte-operation

33. Which method is used to detect double errors:
(a) Odd Parity
(b) Even Parity
(c) Checksum (CRC)
(d) Checksum (XOR)

34. The time delay through an 8-bit serial register with 200 MHz clock is:
(a) 40 ns
(b) 5 ns
(c) 10 ns
(d) 20 ns

35. A full adder circuit requires:
(a) Two inputs and two outputs
(b) Two inputs and three outputs
(c) Three inputs and two outputs
(d) Three inputs and one output

DATABASE SYSTEMS (Questions 36-40)

36. Consider database table T with columns X,Y. After inserting (1,1), records are inserted 64 times with X=MX+1, Y=2*MY+1. What is output of "SELECT Y FROM T WHERE X=5":
(a) 15
(b) 31
(c) 63
(d) 127

37. In SQL, statement "SELECT * FROM R,S" is equivalent to:
(a) SELECT * FROM R NATURAL JOIN S
(b) SELECT * FROM R CROSS JOIN S
(c) SELECT * FROM R OUTER JOIN S
(d) SELECT * FROM R INNER JOIN S

38. Which SQL query deletes teachers from departments in building 'CS':
(a) DELETE FROM teacher WHERE dept_name IN 'CS'
(b) DELETE FROM department WHERE building='CS'
(c) DELETE FROM teacher WHERE dept_name IN (SELECT dept_name FROM department WHERE building='CS')
(d) None of the above

39. Among ACID properties, 'Durability' requires changes persist:
(a) Except in OS crash
(b) Except in disk crash
(c) Except in power failure
(d) Always, even if any failure occurs

40. Fifth Normal Form is concerned with:
(a) Join dependency
(b) Domain-key dependency
(c) Multivalued dependency
(d) Functional dependency

DIGITAL LOGIC (Questions 41-45)

41. Which definition is true:
(i) Multiplexer selects one input from many and outputs selected one
(ii) 1-to-8 demultiplexer requires 3 select lines
(iii) Octal to binary encoder requires 8 OR gates

(a) (i) and (ii)
(b) (ii) and (iii)
(c) (i) alone
(d) All are true

42. The logical operation of NAND gate with inputs tied together:
(a) XOR
(b) AND
(c) OR
(d) NOT

43. Logic circuit providing LOW output when both inputs HIGH or both LOW:
(a) AND
(b) NAND
(c) XOR
(d) XNOR

44. When both inputs of NOR and NAND gates are connected together:
(a) AND function
(b) XOR function
(c) OR function
(d) NOT function

45. For matrix P = [[1,2,3,4],[0,4,3,1],[2,6,4,8],[1,5,2,3]], det|P| is:
(a) Indeterminate
(b) Negative
(c) Zero
(d) Positive

ALGORITHMS & COMPLEXITY (Questions 46-50)

46. Arrange functions in increasing order of complexity: $f_1(n)=3^n$, $f_2(n)=n^{(5/2)}$, $f_3(n)=n^2\log n$, $f_4(n)=n^{(\log n)}$:
(a) f_3, f_2, f_4, f_1
(b) f_1, f_2, f_3, f_4
(c) f_2, f_3, f_4, f_1
(d) f_4, f_3, f_2, f_1

47. Maximum comparisons needed for radix sort of 8 items, each 2-digit decimal:
(a) 80
(b) 160

(c) 200
(d) 240

48. Worst case time complexity of heap sort for n elements:
(a) O(n log n)
(b) O(log n)
(c) $O(n^2)$
(d) O(n)

49. From character string of length m, number of substrings of all lengths:
(a) m^2
(b) m(m+1)/2
(c) m log m
(d) 2^m

50. Complexity of matrix multiplication of matrices with orders m×n and n×p:
(a) O(m × p)
(b) $O(m \times n^2 \times p)$
(c) O(m × n × p)
(d) $O(m \times n \times p^2)$

EMERGING TECHNOLOGIES (Questions 51-55)

51. Which scenario is NOT suitable for HDFS:
(a) Multiple simultaneous writes to same file
(b) Low latency data access applications
(c) High latency data access applications
(d) Large file storage

52. Standard Java API for monitoring applications:
(a) JVM
(b) JMX
(c) JVN
(d) JMY

53. K-means algorithm uses which clustering technique:
(a) Hierarchical
(b) Partitional
(c) Divisive
(d) Agglomerative

54. Private cloud infrastructure is:

(a) For single organization within premises
(b) Shared among multiple organizations
(c) Available to general public
(d) Hybrid of public and private

55. CAPTCHA provides protection from:
(a) Zero day attacks
(b) Buffer overflow
(c) Automated scripted attacks
(d) Man in the middle attacks

ANSWER KEY

1-10: (b), (a), (c), (a), (b), (a), (a), (d), (a), (c)
11-20: (a), (c), (c), (a), (c), (a), (c), (a), (a), (c)
21-30: (a), (d), (b), (b), (a), (a), (c), (a), (b), (b)
31-40: (b), (b), (c), (a), (c), (a), (b), (c), (d), (a)
41-50: (a), (d), (c), (d), (c), (a), (b), (a), (b), (c)
51-55: (a), (b), (b), (a), (c)

Important Notes for Practice:

1. **Time Management**: Attempt these questions in exam conditions - 80 questions in 90 minutes
2. **Focus Areas**: Questions 1-28 are from high-weightage topics in ISRO
3. **Pattern Recognition**: Notice how ISRO tests practical applications of theoretical concepts
4. **Current Trends**: Questions 51-55 reflect ISRO's focus on modern technologies
5. **Calculation Practice**: Questions with numerical computations need step-by-step solving